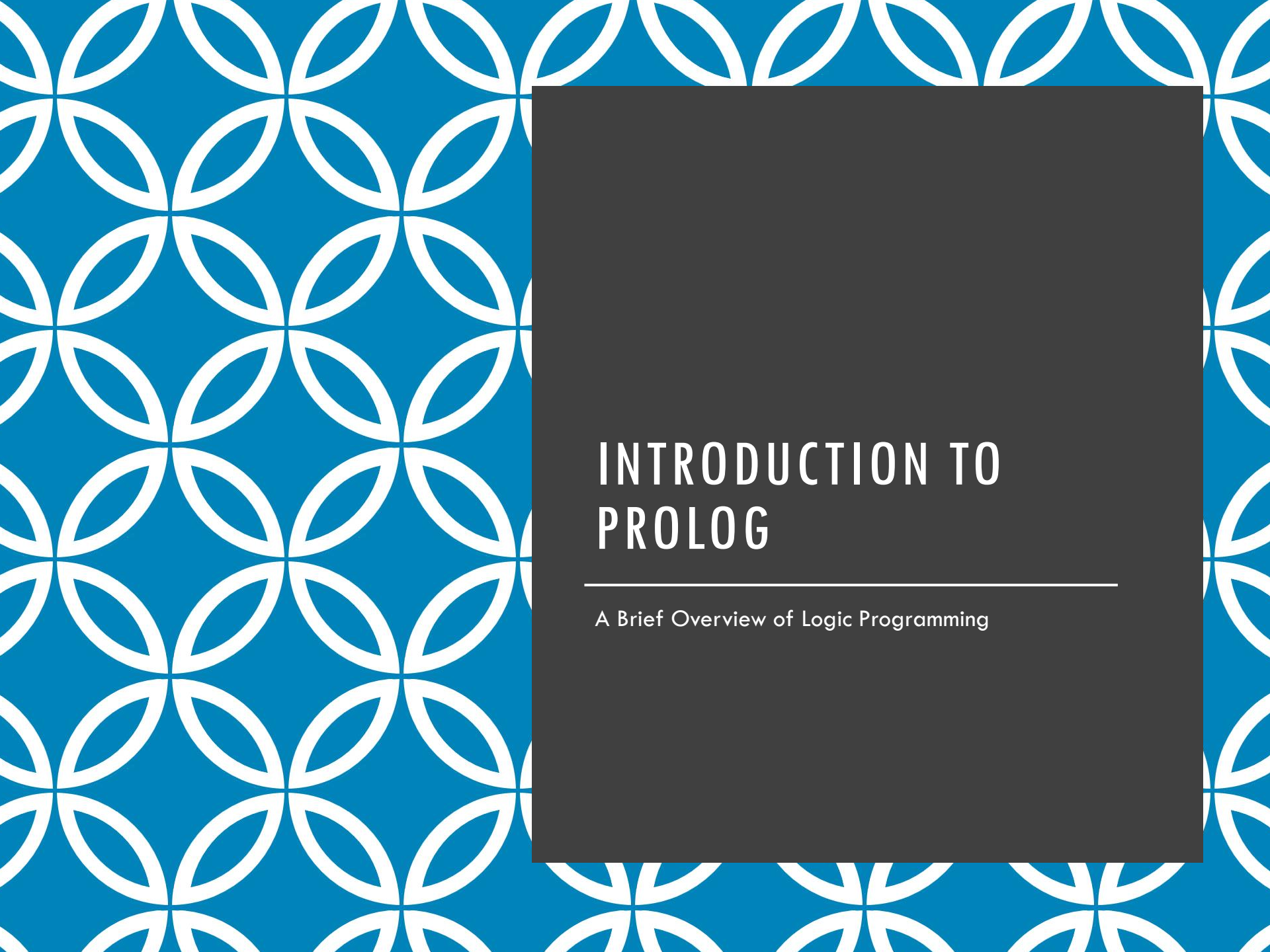




INTRODUCTION TO PROLOG

A Brief Overview of Logic Programming

WHAT IS PROLOG?

Definition: PROLOG (Programming in Logic) is a high-level programming language associated with artificial intelligence and computational linguistics.

Origin: Developed in the early 1970s by Alain Colmerauer and Robert Kowalski.

KEY FEATURES OF PROLOG

Declarative Nature: Focuses on what to solve rather than how to solve it.

Backtracking: Automatically finds all possible solutions.

Pattern Matching: Uses unification for pattern matching.

Recursion: Supports recursive predicates.

BASIC SYNTAX

Facts: Statements that are unconditionally true.

Rules: Conditional statements that define relationships.

Queries: Questions asked to the PROLOG system.

EXAMPLE CODE

Fact Example:

```
parent(tom, bob).
```

Rule Example:

```
grandparent(X, Y) :- parent(X, Z), parent(Z, Y).
```

Query Example:

```
?- grandparent(tom, Y).
```

BASIC ELEMENTS

Facts: These are basic assertions about some information.

`parent(tom, bob).`

`parent(bob, alice)`



% Example facts

male(john).

female(mary).

parent(john, mary).



Rules: These define relationships between facts using logical implications.

`grandparent(X, Y) :- parent(X, Z), parent(Z, Y).`



% Example rule

father(X, Y) :- parent(X, Y), male(X).



Queries: These are questions you ask the PROLOG system to infer information.

?- grandparent(tom, alice).



% Example query

?- father(john, mary).



% Facts

parent(john, mary).

parent(mary, susan).

% Rule

grandparent(X, Y) :- parent(X, Z), parent(Z, Y).

% Query

?- grandparent(john, susan).

`:- initialization(main).`

`main :- write('Hello, World!').`

`% Facts`

`food(burger).`

`food(sandwich).`

`food(pizza).`

`lunch(sandwich).`

`dinner(pizza).`

`% Rule`

`meal(X) :- food(X).`

`% Query`

`?- food(pizza).`

<https://swish.swi-prolog.org/example/kb.pl>

https://www.onlinegdb.com/online_prolog_compiler

TASK

Create a Prolog code for student teacher. Create facts, relationships, rules and queries.

Online Compiler-

<https://swish.swi-prolog.org/example/kb.pl>

APPLICATIONS OF PROLOG

Artificial Intelligence: Natural language processing, expert systems.

Database Systems: Querying and managing databases.

Theorem Proving: Automated reasoning and proof verification.

ADVANTAGES AND DISADVANTAGES

Advantages:

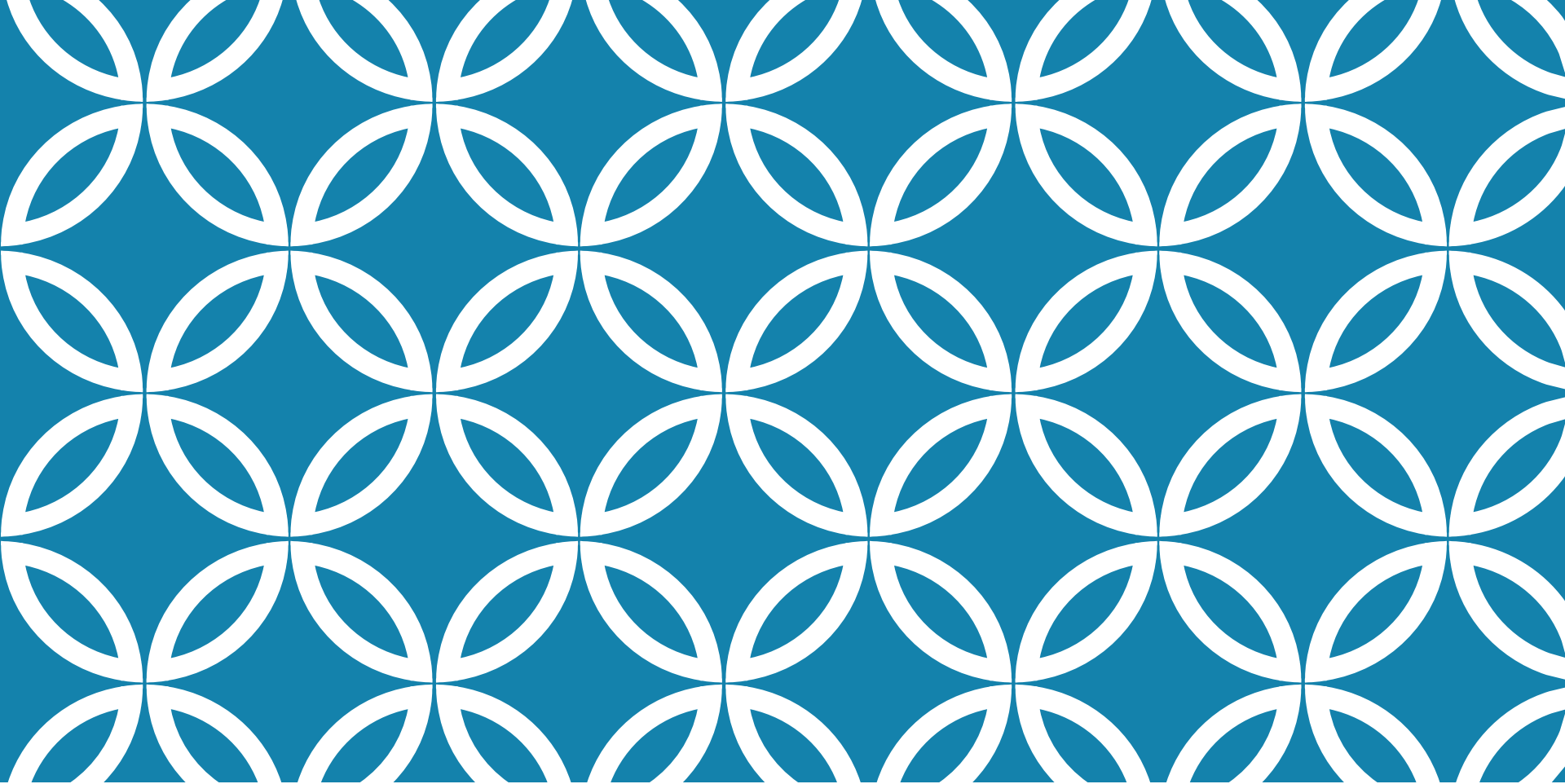
- Easy to express complex relationships.
- Powerful pattern matching and backtracking.

Disadvantages:

- Can be less efficient for certain types of problems.
- Steeper learning curve for beginners.

VIDEO

<https://www.youtube.com/watch?v=Xs6LexZ0BfI>



REASONING UNDER UNCERTAINTY

Understanding and
Applying Uncertainty in
Decision Making



INTRODUCTION

Definition of Uncertainty

Importance of Reasoning Under Uncertainty

UNCERTAINTY

Artificial intelligence (AI) uncertainty is when there's not enough information or ambiguity in data or decision-making. It is a fundamental concept in AI, as real-world data is often noisy and incomplete. AI systems must account for uncertainty to make informed decisions.

<https://intellipaat.com/blog/what-is-uncertainty-in-artificial-intelligence/>

ALEATORY UNCERTAINTY

Aleatory uncertainty, also known as statistical uncertainty, is due to inherent randomness in the system.

Examples:

- Weather conditions
- Stock market fluctuations

EPISTEMIC UNCERTAINTY

Epistemic uncertainty, also known as systematic uncertainty, is due to lack of knowledge or information.

Examples:

- Model parameters
- Measurement errors



METHODS OF REASONING

Probability Theory

Fuzzy Logic

Bayesian Networks

PROBABILITY THEORY

Probability theory is a mathematical framework for quantifying uncertainty.

Key Concepts:

- Random variables
- Probability distributions
- Expected value

FUZZY LOGIC

Fuzzy logic is a form of many-valued logic that deals with approximate reasoning.

Key Concepts:

- Fuzzy sets
- Membership functions
- Fuzzy rules

BAYESIAN NETWORKS

Bayesian networks are graphical models that represent probabilistic relationships among variables.

Key Concepts:

- Nodes and edges
- Conditional probability tables



CASE STUDIES

Practical Examples

Analysis of Outcomes

CASE STUDY 1: WEATHER FORECASTING

Weather forecasting involves reasoning under uncertainty due to the inherent randomness in weather patterns.

Methods Used:

- Probability theory
- Bayesian networks

Outcome:

- Improved accuracy in weather predictions

CASE STUDY 2: MEDICAL DIAGNOSIS

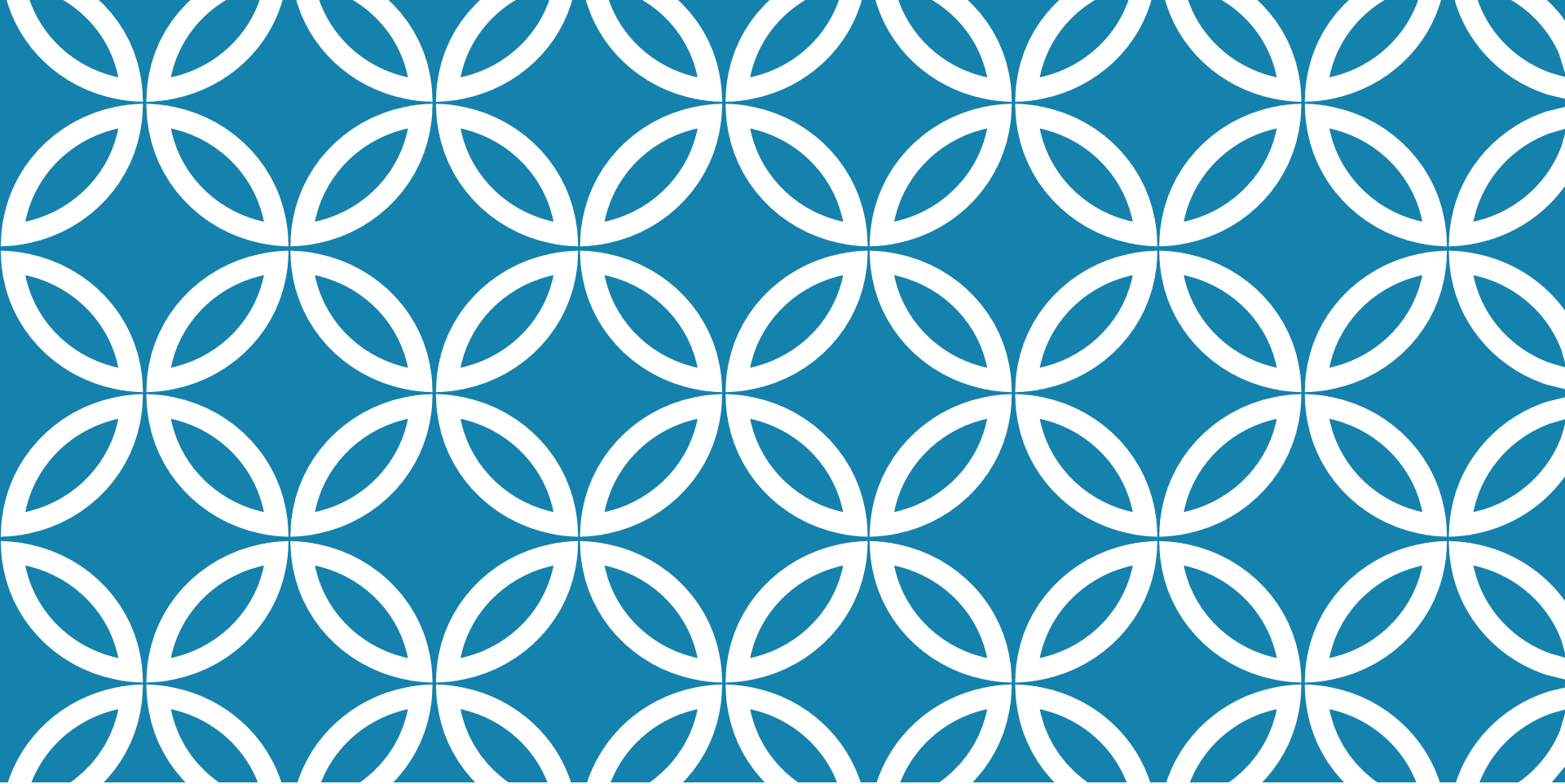
Medical diagnosis involves reasoning under uncertainty due to incomplete knowledge about the patient's condition.

Methods Used:

- Fuzzy logic
- Bayesian networks

Outcome:

- Enhanced decision-making in medical treatments



DATA MINING

Understanding the
Techniques and
Applications of Data
Mining

DATA MINING TECHNIQUES

Data mining is the process of searching and analyzing a large batch of raw data in order to identify patterns and extract useful information.

<https://www.investopedia.com/terms/d/datamining.asp>

CLASSIFICATION

Classification is a data mining technique used to predict group membership for data instances.

Examples:

- Decision Trees

CLUSTERING

Clustering is a data mining technique used to group similar data instances into clusters.

Examples:

- Hierarchical Clustering

ASSOCIATION RULE LEARNING

Association rule learning is a data mining technique used to discover interesting relationships between variables in large datasets.

Examples:

- Growth Algorithm

EMERGING TRENDS

Emerging trends in data mining include:

- Big Data Analytics
- Real-time Data Mining
- Integration with Machine Learning and AI

INDUSTRY-SPECIFIC EXAMPLES

Data mining is used in various industries for different purposes.

Examples:

- Healthcare: Predicting disease outbreaks, patient diagnosis
- Finance: Fraud detection, credit scoring
- Retail: Market basket analysis, customer segmentation

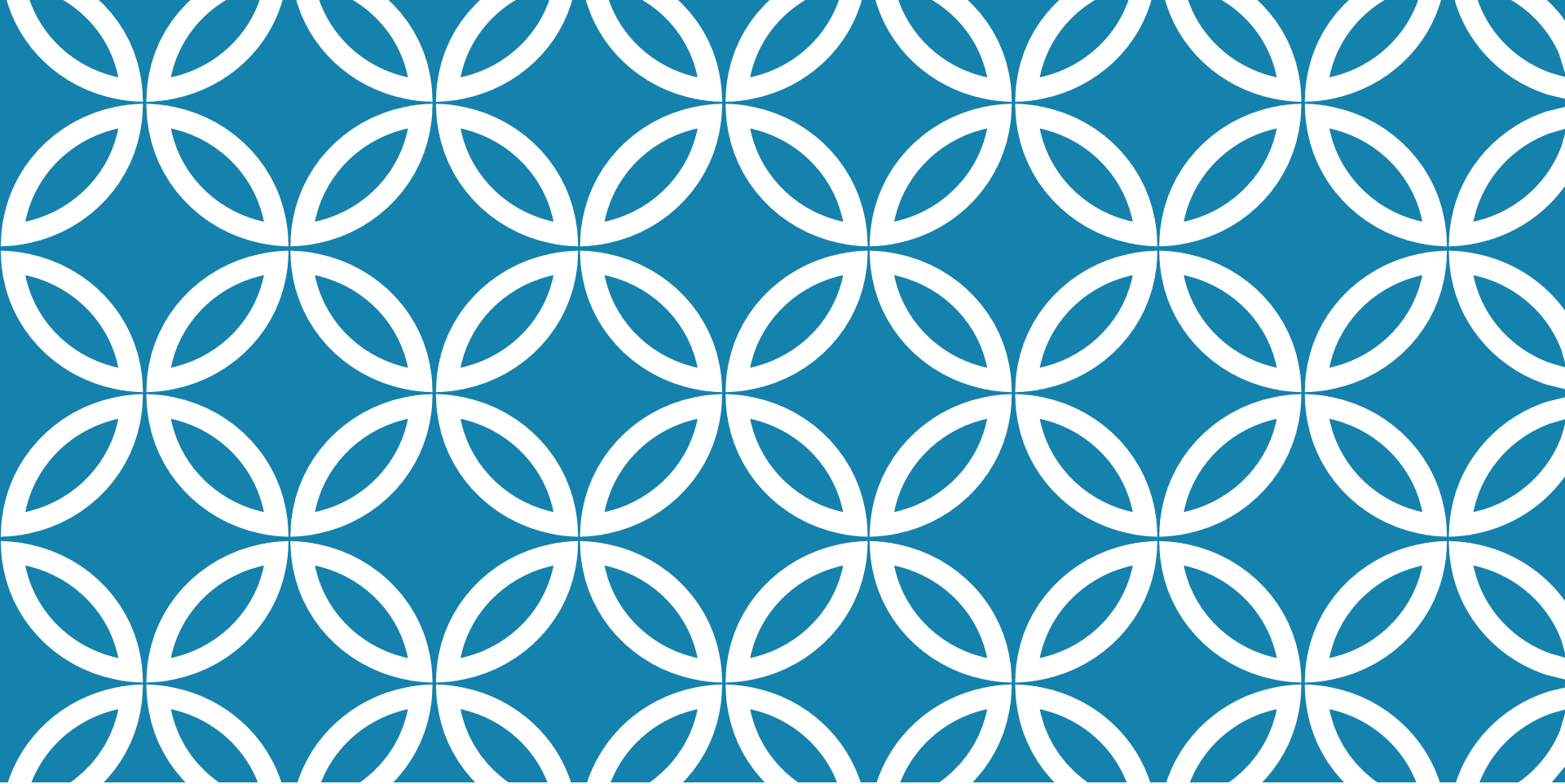
BENEFITS AND CHALLENGES

Benefits of Data Mining:

- Improved decision-making
- Increased efficiency and productivity

Challenges of Data Mining:

- Data privacy and security concerns
- Handling large and complex datasets



NEURAL NETWORKS

Understanding the Basics
and Applications of Neural
Networks

ARCHITECTURE

Neurons and Layers

Types of Neural Networks (e.g., CNN, RNN)



Neural networks are a series of algorithms modeled after the human brain, designed to recognize patterns in data. They interpret sensory data through machine perception, labeling, and clustering raw input. These networks can learn and improve over time, making them essential for tasks like image and speech recognition. This learning path gives you a comprehensive introduction to the topic so you can springboard into more advanced applications for neural networks.

<https://www.pluralsight.com/paths/introduction-to-neural-networks>



TRAINING AND LEARNING

Backpropagation

Optimization Techniques

REAL-WORLD USE CASES

Neural networks are used in various real-world applications.

Examples:

- Image Recognition
- Natural Language Processing
- Autonomous Vehicles

IMPACT ON VARIOUS INDUSTRIES

Neural networks have a significant impact on various industries.

Examples:

- Healthcare: Disease diagnosis, medical imaging
- Finance: Fraud detection, algorithmic trading
- Retail: Customer segmentation, demand forecasting